

Ashton E. Thomas

EDUCATION

University of Michigan - Rackham Graduate School

08/2025 – 05/2026

*Major - MSE Computer Engineering, Computer Vision Concentration**Ann Arbor, MI, USA**summa cum laude, GPA - 4.00/4.00***University of Michigan College of Engineering**

08/2022 – 05/2025

*Major - BSE Computer Science; Minor - International Engineering**Ann Arbor, MI, USA**summa cum laude, GPA - 3.8/4.0**Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics**Data Structures & Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems*

EXPERIENCE

Amazon

09/2024 – Present

*Software Development Engineer I, AWS**Boston, MA & New York, NY, USA*

- Build developer-insights platforms at AWS for software delivery health, observability, and engineering analytics as part of a 12+ person team.
- Completed an SDE internship on Alexa endpoint experiences, shipping features into Alexa+ with a 20+ engineer team using React Native, Kotlin, and TypeScript.
- Managed Amazon company-wide employee data for 100,000s of employees as part of a 10-engineer team, including collection, processing, and executive-level inferences.

Ground Vehicle Systems Center

05/2024 – 07/2024

*Software Development Intern**Warren, MI, USA*

- Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency.
- Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.

Madi Taylor Photo

06/2022 – 07/2024

*Full Stack Developer**Traverse City, MI, USA*

- Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS.
- Implemented robust back-end payment solutions and form validation to streamline user transactions.

RESEARCH

Polk Lab — University of Michigan

09/2024 – 06/2025

*Computational Neuroscience Research Assistant**Ann Arbor, MI, USA*

- Collaborated with a PhD student to develop a machine learning model measuring how different languages affected neural distinctiveness in the human brain, using speech-versus-background-noise separation to support the analysis.
- Worked alongside Prof. Thad Polk to develop a script processing CSV files with 1000s of columns, using wildcard parsing and a node hierarchy to filter, add, remove, print, and interrelate columns.

PROJECTS

Geoguessr AI, Computer Vision

2023 – 2024

- Designed and implemented a modified ResNet-50 architecture for geographic location identification from images.
- Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ~90% accuracy.

Google Search Engine, Web Systems

2024

- Engineered a scalable search engine leveraging a segmented inverted index implemented with MapReduce.
- Integrated tf-idf for text analysis and PageRank for link analysis to improve the relevance of search results produced via a REST API.

Virtual Memory Pager, Operating Systems

2025

- Built a virtual-memory pager managing per-process arenas, page tables, physical frames, swap-backed pages, and shared file-backed mappings.
- Implemented page-fault handling, copy-on-write behavior, dirty/reference tracking, swap allocation, file-block sharing, eviction, and a clock-style page replacement policy.

TECHNICAL SKILLS

ARM, ARCGIS, AWS, Lambda, S3, Glue, Athena, CloudWatch, Redshift, CDK, C#, C++, CI/CD, CSS, Excel, Flask, Git, HTML, Java, Javascript, Jinja, Jira, Keras, Kotlin, Latex, MagicDraw, Matlab, NumPy, Pandas, Python, PyTorch, Qiskit, React/Native, SQL, TensorFlow, Typescript, VSCode